

Assessment 4 Report

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1. Abstract

This report will cover the justifications for the physical design of floor 3 of “Assessment 4 - Network Design (Emerging Technology in IoT)” [1]. Additionally, this report will cover Mason’s own personal reflection on certain aspects of the group assessment. The reflection will be based on Mason’s experience throughout the process of working on “Assessment 4 - Network Design (Emerging Technology in IoT)” [1] as a group. At the end of this report, an understanding of the justification for floor 3’s physical layout along with an understanding of Mason’s group work reflection will be established.

2. Introduction and Objectives

According to the instructions provided in [1], below are the objectives which must be achieved in this report. These objectives are paraphrased from the original instructions provided in the assessment.

Objectives

1. Provide justification for the physical layout of Floor 3.
2. Elaborate upon the tools and techniques learnt throughout the process of conducting the assessment.
3. Demonstrate what the group did well.
4. Demonstrate what the group did not do well.
5. List any issues and how they were resolved.
6. Elaborate upon what was believed to be the most important aspect learnt after the completion of the assessment.

Objective 1 will be an analysis and justification of Mason's physical layout design for Floor 3 [2]. The remaining objectives will be drawn from personal experience.

Definitions

IoT: Internet of Things

PC: Personal Computers

I.D.F: Intermediate Distribution Facility

M.D.F: Main Distribution Facility

3. Results and Discussion

Objective 1

Provide justification for the physical layout of Floor 3. [2]

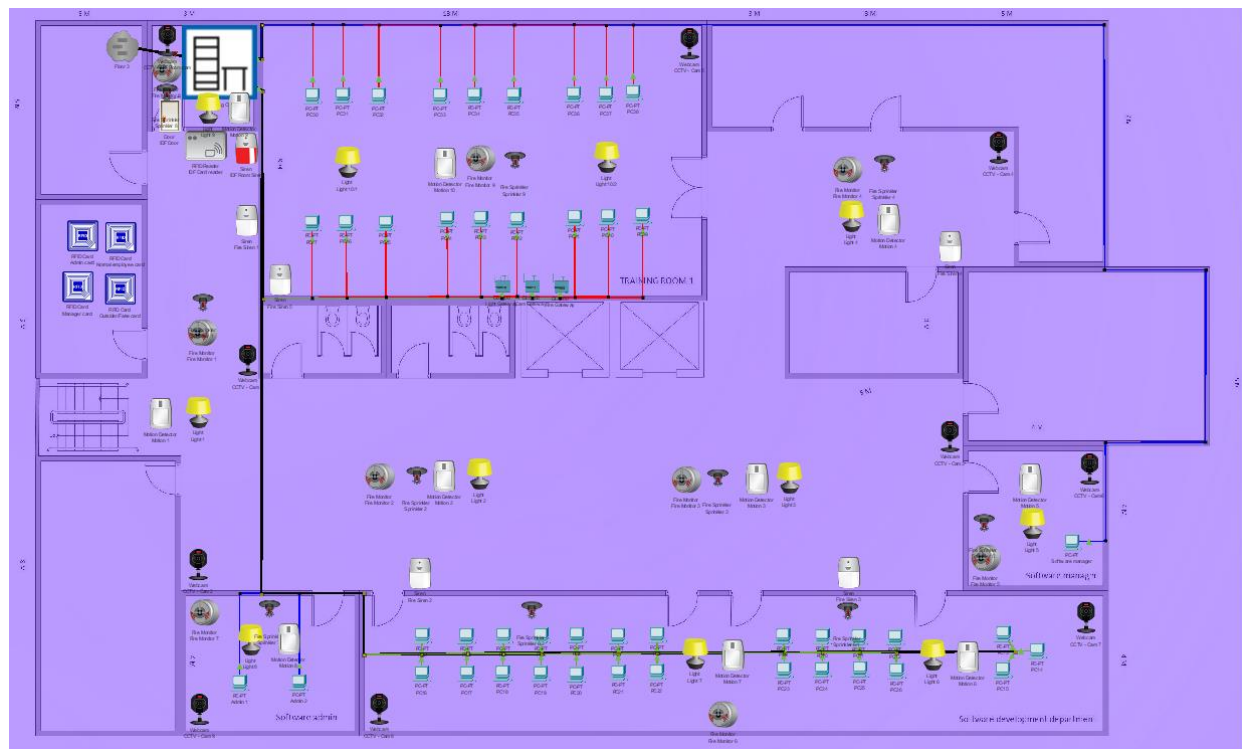


Figure 1: Floor 3's physical layout.

For Floor 3, it was decided to dedicate this floor to the Software Development Department. For starters, this floor will house the first training room for easier access for Software Development trainees. Trainees from the Software Development Department will be given priority to Training Room 1 based on the assumption that most trainees will come from this department; trainees from departments on floors 1 and 2 may still access the training facility if needed. The Software

Development Manager office's will be located between the two staff rooms for easier communication when working or when consultation is desired. The admins for the Software Development Department will share a room located opposite the bathroom to maintain a close vicinity to all the staff and manager for easy communication. Finally, the I.D.F for this level will be located directly above the M.D.F on floor 2 for easier wiring. Additionally, all wires are arranged in a way to minimize visible wires, for better aesthetics.

Regarding the placement of each individual PC. For the training, a more centralized layout was followed to give instructors room to freely move around and provide trainees with instructions. For the placement of Software Development staffs' PC, the devices are arranged into "islands" with employees sitting around on the outside; this was done to allow for freedom of movement and ease of access to all devices. For the administrative devices and the management device, the PCs are arranged in a logical manner with each user facing the respective room's doorway.

Regarding the placement of IoT devices, a general rule followed was to make sure that every corner of the floor was covered by the range of IoT devices. This means that: the fire alarm system covers the entire floor along with all occupied rooms; lighting is available for every occupied area; sirens are hearable no matter where a personnel is on the floor. Cameras are spread out to provide the most coverage for all occupied areas. Except the IDF gateway, which is in the IDF room, all remaining gateways are in the training room to provide sufficient signals for all devices scattered around the floor. All tablets are located inside the IDF room to prevent unauthorized personnel from randomly changing the configuration of IoT devices.

Objective 2

Elaborate upon the tools and techniques learnt throughout the process of conducting the assessment.

Several tools and techniques were acquired after completing the assignment. Many of these relate to specific tools and applications within the Cisco Packet Tracer [3].

Techniques:

- Creation, configuration and placement of IoT devices.
- Wire management in physical view of Cisco Packet Tracer.
- Configuration of Switches, PC, Multiuser.
- Physical placement of PC, Switches.
- Creation and configuration of VLANs.
- Creation of servers.
- IP Addressing.

Tools:

- Reached intermediate level at using Cisco Packet Tracer along with associated applications.
- Usage of When2Meet [4] to organize group activities.

The techniques and tools mentioned above contributed to establishing a sturdy foundation of the usage of Cisco Packet Tracer. Through this assessment, an understanding of network creation and configuration was able to be established.

Objective 3

Demonstrate what the group did well.

Overall, the group had an outstanding performance. First, the communication between all the group members was well beyond expectations; members were actively involved and contributed greatly to the assessment. Second, the workload was fairly divided up with members covering for each other's weaknesses. Finally, group activities were appropriately organized with each member accounted for when deciding on group activities; the usage of When2Meet [4] proved to be extremely effective in communicating and organizing appropriate time for group activities such as team meetings, etc.

Group's Availability

0/4 Available  4/4 Available

Mouseover the Calendar to See Who Is Available

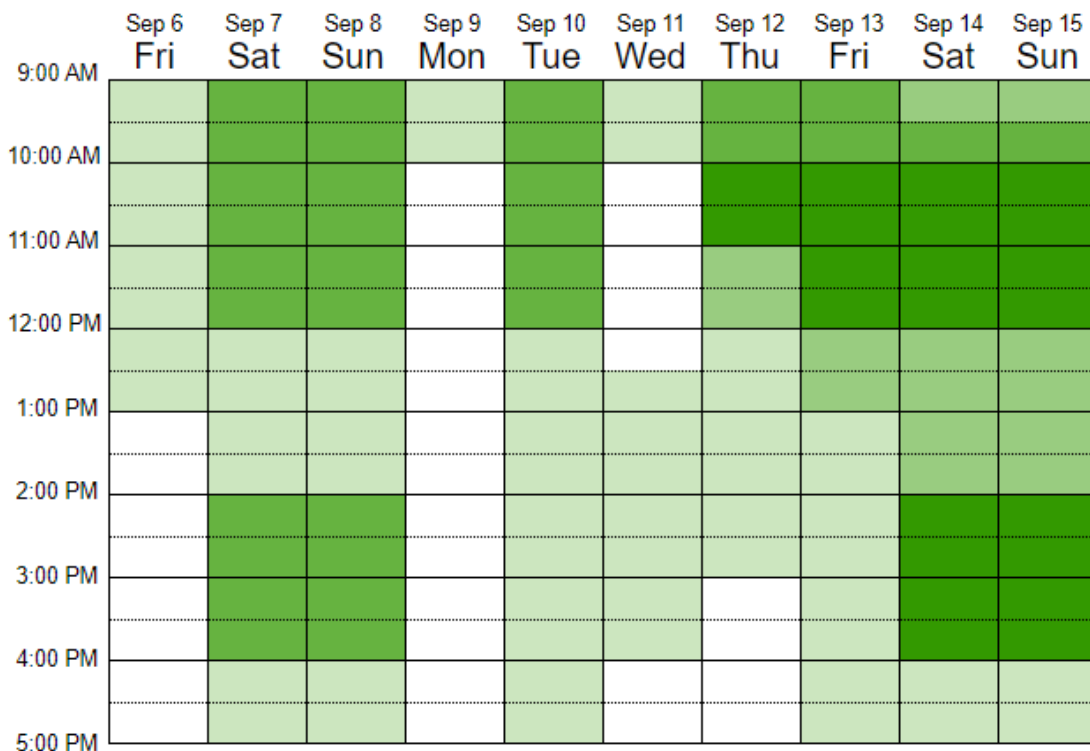


Figure 2: When2Meet table.

Objective 4

Demonstrate what the group did not do well.

One big problem with the group was the delay in starting the assessment. Despite being given a month to prepare and commence work on the assessment, our team members were pre-occupied with assessments from other courses which resulted in a two-week delay in starting work on the assessment. The first real progress was made a bit more than two weeks before the deadline; the progress was on the physical layout of the floors. For the remaining parts of the assessment, progress was halted up into the weekend of week 3 which only left around a bit more than one week before the deadline. This meant there was minimal room for errors, which put some serious pressure on all the group members.

Objective 5

List any issues and how they were resolved.

There were quite a few issues which arose during the process of working on the assessment.

First, IP addressing part of the assessment was a big challenge as our understanding of the networking system was lacking, which resulted in misconfigurations of the network at first. To fix this issue, our group conducted a meeting where we sat down together to discuss problems and come up with a solution. This meeting lasted 6 hours but we were able to resolve the issue in the end.

Second, when planning and configuring the IoT devices, Cisco Packet Tracer [3] is extremely buggy which causes devices to randomly stop working or not connected properly. To combat this issue, our team had to sit through countless hours, patiently adjusting and reconfiguring until all the devices worked properly.

Finally, there were a few minor hiccups during the configuration and setup of the switches,

routers, VLANs and the multiuser. However, we were able to quickly address and solve them through doing a few research and communicating with each other through text messages.

Objective 6

Elaborate upon what was believed to be the most important aspect learnt after the completion of the assessment.

The most important aspect arguably is the skill and knowledge of operating Cisco Packet Tracer. Cisco Packet Tracer is a widely known and used tool which is often closely associated with the Data Communication field of Information Technology. Possessing such important fundamentals on the application before stepping foot into the real world is essential for future self-development. Cisco Packet Tracer does an excellent job at demonstrating real world principles in a controlled and educational environment, which helps not overwhelm newcomers; additionally, Cisco Packet Tracer is a simple to use along with providing much flexibility and customizability.

4. Conclusion

In conclusion, this assessment was an insightful experience, which covers many areas of Cisco Packet Tracer. Through this experiment, I was able to successfully set up and configure a simulated network with realistic Data Communications systems. Much insight into the varying aspects of IoT and multiuser was also touched upon. Furthermore, I was able to observe some of the fundamental protocols of the application layer in working. Finally, my skills working in a group environment have also developed considerably, along with gaining new knowledge on useful tools which may aid me in the future.

References

- [1] RMIT University. "Assessment 4 - Network Design (Emerging Technology in IoT)". RMIT Instructure. (Accessed: Sep 18, 2024). [Online]. Available: <https://rmit.instructure.com/courses/138109/assignments/958071>.
- [2] M. Mason. "Assessment 4 - Packet Tracer Submission", "Floor3_G1_Mason". Unpublished. Sep 2024.
- [3] Cisco Packet Tracer, version 8.2.2.0400, Cisco Systems, Inc., 2024.
- [4] When2Meet. [Online]. Available: <https://www.when2meet.com/>.